

Arrays

Simulate the following array-based problems using C language:

1. Given an array $A[1...N]$ of integers and N , the number of elements in the array, retrieve the maximum and minimum element using brute force technique.
 - a. Find the largest element of the array A for Q1, using the tournament method.
Reference: <https://users.csc.calpoly.edu/~dekhtyar/349-Fall2017/lectures/lec03.349.pdf>
 - b. Find the second largest element of array A using the tournament method with the help of the solution you obtained for Q1.a.
 - c. Simulate Dynamic Memory Allocation technique with the help of predefined functions and print the array elements.
2. Write a program to calculate and print the sum of all elements in a given array using both **iterative** and **recursive** technique.
3. Implement linear search and count the occurrence of the given element x .
4. Consider a 2D array $A[i][j]$, print all the elements by navigating using a spider search.
5. During a disaster relief operation, you're piloting a helicopter that must visit a series of islands lined up in a straight path. The number of people stranded on each island is recorded in an array. However, due to safety and landing pad design, your helicopter is only certified to land on every alternate island starting from the second—i.e., the odd-numbered positions when counted from 1 (1-based index: 1, 3, 5...). Each landing allows you to rescue all people from that island. Your task is to determine the total number of people you can rescue by landing only on the permitted islands.
Example: Input: $A = [10, 20, 30, 40, 50, 60]$
Output: 120 (*Rescue from positions 1, 3, 5 $\rightarrow 20 + 40 + 60 = 120$*)
6. Two adventurers stumble upon a mystical sequence of coins. The magic of the treasure requires that the coins be divided exactly in the middle — into two non-overlapping, contiguous parts — such that both parts have equal total value. You are given an array of integers representing the values of the coins in a sequence. Determine whether such a fair split is possible. If yes, the adventurers can each take their share in peace. If not, the magical balance is broken.
Example: Input: $A = [10, 20, 30, 30, 20, 10]$
Output: Yes (*Split into $[10, 20, 30]$ and $[30, 20, 10]$, both sum to 60*)
7. In the heart of a mystical arena, a circular shield of warriors is formed, each occupying a unique position in a ring-shaped structure. The positions of the warriors are stored in an array. Before every battle, the shield spins exactly k times to the right due to a magical command. This spinning operation shifts each warrior k positions forward, with the last ones wrapping around to the front. Your task is to simulate this rotation and output the new arrangement of the warriors after k rightward shifts using both **iterative** and **recursive** method.
Example: Input: $A = [1, 2, 3, 4, 5]$, $k = 2$;
Output: $[4, 5, 1, 2, 3]$ (*After two right rotations*)

References:

https://www.cs.sfu.ca/~ashriram/Courses/CS295/assets/books/C_Book_2nd.pdf

https://oku.ozturkibrahim.com/clab/booksEN/C_How_to_Program_7th_Ed.pdf

Moodle Submission Logistics:

- You need to have the following files in a zipped folder with the naming format:
"Arrays_<<roll_no>>"
 - (a) Your code in .C format along with the outcome (within the comment) with proper Q.No. and Question name.
 - (b) any reference materials you have used arranged in a folder.

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